

Undominated Voting in the Clean Immediate-Opt-Out Baseline

A Systemic Rederivation under Simultaneous Ballots

Goal 3 formal workspace

2026-08-04

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1 Status, scope, and provenance

This is the autonomous formal workspace for Goal 3. It does not amend or compile the manuscript. Its purpose is to solve the clean two-round game after replacing the old ballot discipline with **PBE plus local weak-undominance of every vote used at every ballot information set** (PBE-UD).

The input snapshot is the annotated tag **pre-undominated-voting-repair-2026-08-04**, which points to commit **8692a6b31539d117134a580009d4e4dbe783403d**. These artifacts are frozen:

```
formal_model_v6.Rmd
131cc2356cd6318211fdbb9304ac8d7c8a99356837b6e71097011c36ae9c270d

formal_model_v6.pdf
a7d36d5a1fb2d15ba0e40509ad846fbf001b5960232fcfb011d5b32bec298bdf

model_redesign/power_architecture_derivations.Rmd
8fbb7edff59fb0dc6fb36571564ec94d26e66b1211496ed10e9d9191ef2f68c6
```

The post-closure erratum is **quality_reports/2026-08-04_goal2_postclosure_bf_voting_erratum.md**; the autonomous voting contract is **quality_reports/2026-08-04_undominated_voting_protocol_gate0.md**. Gate 0 passed two independent reviews, was reopened after a terminal stress test exposed an overbroad threshold corollary, and passed rereview after repair.

1.1 Result-status convention

Every result is marked **proved**, **checked numerically**, **conjecture**, **pending**, or **rejected**. Enumeration is a regression check, never a substitute for proof.

2 Fixed game contract

There are $N \geq 3$ states: a hegemon H and $m = N - 1$ weak states. Nature draws $\theta \in \{0, 1\}$, observed only by H , with $\mu = \Pr(\theta = 1)$. The primitive domain is

$$0 \leq o_0 < o_1 \leq \bar{y} \leq 1, \quad \beta \in (0, 1]. \quad (1)$$

The weak institutional pie is one and does not depend on type. Weak outside payoffs are zero. The payoff o_θ is external to that pie. The baseline fixes $\pi_H = 0$ and $b_\theta = 0$ in both rounds. A current agreement that includes H pays it y .

The exogenous rule is $R \in \{U, M\}$, with quotas

$$Q_U = N, \quad Q_M = q = \left\lfloor \frac{N}{2} \right\rfloor + 1. \quad (2)$$

The proposer is counted as voting yes. All other active players vote simultaneously without observing any other ballot. The complete vote vector and outcome become public only after closure. There is no within-ballot order, no roll call, and no position of H over which to conduct order sensitivity.

A no vote by type θ of H in R1 pays o_θ immediately and irreversibly. Under unanimity that ends the branch. Under majority, a current weak-only agreement passes if weak yes votes reach q ; otherwise R2 is a weak-only continuation. If H votes yes but weak votes cause failure, H remains active and R2 follows. R2 is terminal.

2.1 Proposals and budgets

When weak state i proposes while H is active,

$$s_i = (y, (x_j)_{j \neq i}), \quad y \in [0, \bar{y}], \quad x_j \geq 0. \quad (3)$$

If H participates, the proposer's residual is

$$x_i^H = 1 - y - \sum_{j \neq i} x_j \geq 0. \quad (4)$$

If H votes no and a majority weak-only agreement passes, y is reabsorbed:

$$x_i^{WO} = 1 - \sum_{j \neq i} x_j. \quad (5)$$

After opt-out, R2 proposals fix $y = 0$. A named offer is paid whenever the current proposal passes, even if its recipient voted no.

2.2 Formation

Before R1 recognition, the m weak states make a collective all-or-nothing formation decision and each pays the external sunk cost χ . For a selected equilibrium under rule R , let T_W^R be expected total gross weak payoff. Under anonymous recognition and symmetric partner selection,

$$V_W^R = \frac{T_W^R}{m}, \quad \text{formation iff } V_W^R \geq \chi. \quad (6)$$

When asymmetric transfers are allowed, T_W^R/m remains the collective average used by the formation primitive; it is not an individual entry claim for every realized coalition partner.

3 Solution concept

Fix a candidate continuation assessment κ . The same κ is used to evaluate both votes against every feasible pure profile of the other voters. Beliefs and continuation strategies are not re-solved separately for each counterfactual profile. They must recursively satisfy PBE-UD.

For an uninformed weak voter j , let $\Omega_{-j}(I)$ contain all feasible pure plans of the other weak voters and all type-contingent plans of H . Other weak actions cannot condition on θ . Define

$$\Delta_j(\omega_{-j} \mid I, \kappa) = \sum_{\theta} \nu_I(\theta) [u_j(Y, \omega_{-j}, \theta \mid I, \kappa) - u_j(N, \omega_{-j}, \theta \mid I, \kappa)]. \quad (7)$$

Vote Y is weakly dominated by N if $\Delta_j \leq 0$ for every ω_{-j} , with a strict inequality for at least one profile. Reverse inequalities define domination of N . This is one-shot interim Bayesian deletion, not iterative deletion, trembling-hand perfection, or an “as if pivotal” rule.

For H , the comparison is type-specific:

$$\Delta_H(\theta, v_W \mid I, \kappa) = u_H(Y, v_W; \theta, I, \kappa) - u_H(N, v_W; \theta, I, \kappa), \quad (8)$$

where v_W is a counterfactual weak vector, not an observed vector. A PBE-UD is a PBE whose behavioral ballot strategy places positive probability only on locally undominated votes.

3.1 Equality objects

The primary object leaves both votes available when neither dominates. Three selections are then reported:

1. T^Y : yes at every voting indifference;
2. T^N : no at every voting indifference;
3. T^p : an explicitly stated behavioral mixture.

Proposer indifference is separate. Among proposal payoff ties, the legacy selection minimizes H 's expected payoff; the opposite selection maximizes it; without selection, the full correspondence remains.

4 Local voting lemmas

Lemma 1 (proved: terminal weak vote). At a reachable terminal ballot, $x_j > 0$ makes N weakly dominated by Y . At $x_j = 0$, the two votes are payoff-equivalent in every feasible profile and both remain admissible.

Proof. Nonpivotal profiles give the same payoff under both actions. In a pivotal profile, Y implements x_j and N produces terminal zero. Every reachable quota admits such a profile. $\square$

Lemma 2 (proved: terminal H vote). With H active in R2,

$$y < o_\theta \Rightarrow UD_H(\theta) = \{N\}, \quad y > o_\theta \Rightarrow UD_H(\theta) = \{Y\}, \quad (9)$$

while $y = o_\theta$ leaves both votes admissible.

Lemma 3 (proved, conditional scalar reduction). Suppose a weak voter's actions differ only in pivotal profiles; pivotal yes implements x_j ; pivotal no always produces the same continuation c_j ; and a pivotal profile is feasible. Then

$$x_j < c_j \Rightarrow UD_j = \{N\}, \quad x_j > c_j \Rightarrow UD_j = \{Y\}, \quad (10)$$

and equality leaves both votes admissible. This lemma is not used if public vote vectors induce different continuation values.

For an R1 vote by H , let $V_\theta(v_W)$ equal y after current implementation and $\beta C_{H,2}^R(\theta, h_2(Y, v_W))$ after weak-caused failure.

Lemma 4 (proved: R1 H min-max test). Vote Y is dominated by opt-out exactly when

$$\max_{v_W} V_\theta(v_W) \leq o_\theta \quad \text{and} \quad \min_{v_W} V_\theta(v_W) < o_\theta. \quad (11)$$

The reverse weak inequalities, with at least one strict, characterize domination of opt-out. Otherwise both votes are locally admissible and PBE expected incentives decide their use.

An exact offer $y = o_0$ is not automatically rejected: an R2 pooling continuation can pay the low type o_1 , and βo_1 may exceed o_0 . Under majority with $N = 3$, proposer plus $H = Y$ already attain $q = 2$, so no weak-caused failure vector exists and both H votes remain admissible at $y = o_\theta$.

5 Round 2

Define

$$D = 1 - o_0, \quad P = 1 - o_1, \quad L(\nu) = (1 - \nu)D, \quad S(\nu) = \max\{L(\nu), P\}. \quad (12)$$

5.1 Unanimity with H active

Proposition 1 (proved, conditional value). Every terminal unanimity PBE-UD, if one exists, gives the recognized proposer $S(\nu)$ and each weak state ex ante value $S(\nu)/m$. Under T^Y , low-only at $y = o_0, x = 0$ is selected when $L(\nu) > P$, pooling at $y = o_1, x = 0$ when $P > L(\nu)$, and both classes are available at equality. The cutoff is

$$\nu_2^* = \frac{o_1 - o_0}{1 - o_0}. \quad (13)$$

Low-only gives H payoffs (o_0, o_1) ; pooling gives (o_1, o_1) . At $L = P$, the proposal tie-break minimizing H selects low-only, the opposite selects pooling, and no selection retains both and their mixtures.

Attainment. If $\bar{y} > o_1$, a best response exists only if a maximizing boundary is accepted with probability one by the marginal type of H and by every required zero-paid weak voter. If the maximizing boundary is not attained, strict offers approach $S(\nu)$ but no pure or mixed proposal attains it.

At $\bar{y} = o_1$, let λ_1 be high-type acceptance at the cap. The cap proposal gives

$$J(\lambda_1) = \{1 - \nu + \nu\lambda_1\}P. \quad (14)$$

It is optimal if $J(\lambda_1) \geq L(\nu)$; otherwise the low boundary must be attained. Thus the value is unique conditional on existence, but existence is not selection-free.

5.2 Majority

Let $k = q - 1$, the number of nonproposing weak yes votes required when H does not count.

Proposition 2 (proved). A terminal majority PBE-UD exists if and only if at least k zero-paid nonproposers vote yes with probability one. Every such equilibrium gives

$$C_{W,2}^M = \frac{1}{m}, \quad C_{H,2}^M(\theta) = o_\theta, \quad (15)$$

with proposer payoff one and total weak payoff one. The same result applies after R1 opt-out, except that H receives no new R2 flow.

Proof. Paying $\varepsilon > 0$ to any k weak voters forces their yes votes and gives the proposer $1 - k\varepsilon$. Hence a best response must attain one. Attainment requires zero total offers and sure weak-only passage, which is possible exactly with a free winning support of k voters. $\square$

Under T^Y this support exists. Under global T^N , or a proper mix that leaves positive probability of missing the quota, no pure or mixed proposer best response exists. Mixing cannot attain a supremum that no pure action in its support attains.

Table 1: Terminal PBE-UD existence and equality sensitivity.

Terminal subgame	Full correspondence / T^Y	Global T^N	Proper mixing at every required equality
U, H active	Value $S(\nu)$ if a maximizing boundary is attained	no PBE	no PBE unless one maximizing boundary is accepted surely
M, H active	weak-only passage; proposer one	no PBE	no PBE unless k zero-paid voters approve surely
M, post-opt-out	weak-only passage; proposer one	no PBE	same sure-support condition

6 Round 1 under majority

This section uses the regular interior domain

$$\mu \in (0, 1), \quad \beta \in (0, 1), \quad 0 < o_0 < o_1 < 1. \quad (16)$$

Write

$$k = q - 1, \quad r = q - 2, \quad c = \frac{\beta}{m}, \quad (17)$$

and define the three relevant proposer limits

$$\begin{aligned} E &= 1 - kc, \\ B &= (1 - \mu)\{1 - o_0 - rc\} + \mu c, \\ C &= 1 - o_1 - rc. \end{aligned} \quad (18)$$

Here E is attained exclusion; B is the low-only supremum; and C is the pooling supremum, active only if $\bar{y} > o_1$.

6.1 Voting reduction

Lemma 5 (proved). In every R1 majority subgame whose R2 continuation satisfies Proposition 2,

$$x_j < c \Rightarrow UD_j = \{N\}, \quad x_j > c \Rightarrow UD_j = \{Y\}, \quad (19)$$

and both votes are admissible at $x_j = c$.

Proof. A pivotal no vote leads either to active- H R2 or post-opt-out weak-only R2. Proposition 2 gives the same discounted weak continuation c in both cases. Lemma 3 applies. $\square$

For $N \geq 4$, a weak-caused failure vector exists after $H = Y$, and R2 majority pays type θ only $\beta o_\theta < o_\theta$. Hence

$$y = o_\theta \implies H_\theta = Y \text{ is weakly dominated.} \quad (20)$$

For $y > o_\theta$, both votes are locally admissible. If p is the equilibrium probability that $H = Y$ implements,

$$H_\theta = Y \iff p \geq \frac{(1 - \beta)o_\theta}{y - \beta o_\theta}, \quad (21)$$

with equality permitting mixing. Equation (20) is the dominance restriction; equation (21) is sequential rationality.

6.2 $N \geq 4$

With s c-paid weak supporters:

- $s \leq q - 3$: even $H = Y$ cannot implement;
- $s = q - 2 = r$: implementation requires $H = Y$;
- $s \geq q - 1 = k$: the weak coalition passes without H .

This derives the support size; it does not impose a minimum winning coalition as a primitive.

Theorem 1 (proved: regular majority, $N \geq 4$). Suppose every reached terminal majority subgame has a free winning support, and suppose an R1 exclusion coalition of k voters offered c approves with probability one. A pure or mixed PBE-UD exists if and only if

$$E \geq B \quad \text{and} \quad [\bar{y} = o_1 \text{ or } E \geq C]. \quad (22)$$

When it exists, every equilibrium is payoff-equivalent to exclusion:

$$T_W^M = 1, \quad V_W^M = \frac{1}{m}, \quad u_H^M(\theta) = o_\theta. \quad (23)$$

The recognized proposer obtains E . Multiplicity concerns coalition identity, payoff-irrelevant offers $y < o_0$ to the excluded hegemon, and mixtures over payoff-equivalent exclusion proposals, not the payoff vector. An additional supporter paid c is not payoff redundant: it strictly lowers the proposer residual and therefore cannot appear in an E -attaining best response.

Proof. Exclusion with k offers of c attains E . A strict low-only proposal uses r supporters and approaches

$$(1 - \mu)\{1 - o_0 - rc\} + \mu c = B$$

as $y \downarrow o_0$, but cannot attain it because (20) removes low-type yes at $y = o_0$. If $\bar{y} > o_1$, strict pooling with r supporters approaches C as $y \downarrow o_1$, but cannot attain it because high-type yes is removed at the boundary. An oversized separating proposal with k supporters gives

$$E - (1 - \mu)y < E$$

and is strictly dominated for the proposer by exclusion with the same support. More supporters only increase cost.

To exhaust mixed weak and hegemonic votes, first note that every weak voter who votes yes with positive probability must receive at least c . Offers strictly above c induce the same sure yes as c and strictly reduce the current residual. Offers at c may mix; conditional on each realized weak vote profile and each type-contingent vote of H , the proposer obtains a convex combination of: continuation, an r -support outcome requiring $H = Y$, and a k -support weak-only outcome. The first two are bounded by the low-only or pooling limits B and C , according to which types of H approve; the last is bounded by E . Randomization by H or by threshold-paid weak voters only convexifies these conditional values and cannot exceed $\max\{E, B, C\}$. Likewise, a proposer mixture can be optimal only if every pure proposal in its support attains the same maximum.

It follows that E is a best response exactly under (22). If either strict supremum exceeds E , no pure action attains the global supremum, and neither voter mixing nor proposer mixing can repair the empty best-response set. When E is maximal, any best response that attains it uses exactly k threshold-paid sure yes votes and excludes H , yielding (23). $\square$

The conditions simplify to

$$E \geq C \iff o_1 \geq c, \tag{24}$$

and

$$E \geq B \iff \begin{cases} \text{always,} & o_0 \geq c, \\ \mu \geq \mu_M := \frac{c - o_0}{E - o_0}, & o_0 < c. \end{cases} \tag{25}$$

The parameter condition is the same for $N = 4$ and $N \geq 5$, but the support geometry differs. At $N = 4$, all two nonproposing weak states are needed for exclusion. At $N \geq 5$, there is slack in coalition identity, but not in the total cost kc . Therefore the old claim that large groups always restore existence is **rejected**.

For the required diagnostic $N = 5, q = 3$, including H needs one weak supporter and has candidate boundary cost $o_1 + c$; excluding H needs two weak supporters and costs $2c$. These expressions emerge from (19).

6.3 $N = 3$

Now $m = 2, q = 2, k = 1, r = 0, c = \beta/2$. Proposer plus $H = Y$ always pass, so the exact H thresholds remain admissible. Define

$$\begin{aligned}
E &= 1 - c, \\
B &= (1 - \mu)(1 - o_0) + \mu c, \\
C &= 1 - o_1, \\
Z &= (1 - \mu)C + \mu c.
\end{aligned} \tag{26}$$

Let λ_0 be low-type acceptance at $y = o_0$, λ_1 high-type acceptance at $y = o_1$, and let $e = 1$ when an exclusion proposal at $x = c$ is supported by a sure weak yes. Exact-threshold payoffs are

$$Q_0(\lambda_0) = (1 - \lambda_0)c + \lambda_0 B, \quad Q_1(\lambda_1) = Z + \lambda_1(C - Z). \tag{27}$$

Theorem 2 (proved: regular majority, $N = 3$). Suppose every reached terminal majority subgame has a valid free winning support.

If $\bar{y} > o_1$, set $V = \max\{E, B, C\}$. A PBE-UD exists under a fixed equality assessment if and only if

$$\{E = V \text{ and } e = 1\} \quad \text{or} \quad \{B = V \text{ and } \lambda_0 = 1\} \quad \text{or} \quad \{C = V \text{ and } \lambda_1 = 1\}. \tag{28}$$

Partial threshold acceptance never exceeds V and cannot rescue an unattained strict maximum. Under T^Y , every class is attained and proposer payoff is $\max\{E, B, C\}$.

If $\bar{y} = o_1$, strict pooling is unavailable. Set

$$V_{\text{cap}} = \max\{E, B, Q_1(\lambda_1)\}. \tag{29}$$

Existence requires an attained maximizer: $E = V_{\text{cap}}$ with $e = 1$, $B = V_{\text{cap}}$ with $\lambda_0 = 1$, or $Q_1(\lambda_1) = V_{\text{cap}}$. The cap class is attained even under partial high-type acceptance and can be uniquely optimal.

Proof. Let x be the offer to the sole nonproposing weak voter. If $x < c$, that voter chooses no, and the proposer maximizes within this region at $x = 0$. If $x = c$, the weak voter may mix; if $x > c$, the voter chooses yes surely, but any excess above c strictly reduces the proposer residual. Thus the only candidate values are exclusion, low-only, pooling, and their exact-threshold mixtures in (27).

Because $c < E$,

$$Q_0(\lambda_0) = (1 - \lambda_0)c + \lambda_0 B \leq \max\{E, B\}.$$

Moreover,

$$Z = B - (1 - \mu)(o_1 - o_0) < B,$$

so $Q_1(\lambda_1) \leq \max\{B, C\}$. Therefore, away from the cap, partial threshold acceptance cannot exceed V ; a maximizing class is a best response exactly when its required equality action is actually attained, which gives (28). At the cap, the actual class $Q_1(\lambda_1)$ replaces the unavailable strict-pooling value C and is attained for every fixed λ_1 . The same comparison yields (29) and its attainment test. A mixed proposer strategy adds only mixtures over pure actions that already attain the common maximum. $\square$

Table 2: Regular $N = 3$ majority payoff correspondence.

$N = 3$ class	Total weak payoff T_W^M	Collective average $T_W^M/2$	Expected H payoff
Exclusion E	1	1/2	$\bar{o}$
Low-only B	$(1 - \mu)(1 - o_0) + \mu\beta$	total divided by 2	$\bar{o}$
Pooling C	P		o_1

$N = 3$ class	Total weak payoff T_W^M	Collective average $T_W^M/2$	Expected H payoff
Cap Q_1	$(1 - \mu)P + \mu\{\lambda_1 P + (1 - \lambda_1)\beta\}$	total divided by 2	o_1

At a proposer tie, minimizing H 's payoff selects exclusion or low-only over pooling/cap; maximizing H reverses that choice. An $E = B$ tie remains set-valued because both give $H \bar{o}$. Without selection, all attained maximizers and mixtures over them remain.

6.4 Majority equality sensitivity

Table 3: Majority PBE-UD sensitivity to voter equality behavior.

Selection	Terminal requirement	Regular $N \geq 4$	Regular $N = 3$
T^Y	free terminal support exists	Theorem 1 conditions; exclusion	always exists; $\max\{E, B, C\}$
global T^N	terminal best response absent	no full PBE	no full PBE
proper global T^P	terminal support misses quota with positive probability	no full PBE	no full PBE
R1-only mixing with valid R2	threshold supporters must still pass surely	payoff-equivalent exclusion if Theorem 1 holds	Theorem 2 attainment test

7 Round 1 under unanimity

Continue on the regular domain (16). Define

$$\delta = \frac{\beta(m-1)}{m}, \quad a = 1 - \delta, \quad \nu_2^* = \frac{o_1 - o_0}{1 - o_0}. \quad (30)$$

The exact R1 pooling package $y = o_1$ is no longer admissible: after any weak-caused failure, the high type receives only $\beta o_1 < o_1$, whereas current implementation pays exactly o_1 . Thus high-type yes is weakly dominated. On a separating path, $H = Y$ reveals the low type, terminal unanimity selects low-only, and $y = o_0$ similarly makes low-type yes weakly dominated. These two observations reject the canonical exact-threshold R1 packages.

7.1 Security floor and weak-PBE completion

When $\bar{y} > o_1$, define

$$K(\mu) = \max\{a(1 - \mu)D, P - \delta D\}. \quad (31)$$

The first component is approached by strict low-only proposals $y \downarrow o_0$ that pay each weak voter just above $\beta D/m$. The second is approached by strict pooling proposals $y \downarrow o_1$ with the same punitive weak continuation. At the cap $\bar{y} = o_1$, the second deviation is unavailable and the floor is only $a(1 - \mu)D$.

Lemma 6 (proved: unanimity security and completion). Every regular unanimity PBE-UD gives the recognized proposer at least the applicable floor in (31). Conversely, an on-path candidate satisfying Bayes' rule, all vote ICs, local PBE-UD, budget feasibility, and proposer payoff at least $K(\mu)$ can be completed as a weak PBE-UD when $\bar{y} > o_1$.

Completion. At every globally off-path proposal, assign probability one to the low type both at its ballot and after every $H = Y$, weak-failure history. Select terminal low-only and yes at its required equalities. This gives

weak continuation $\beta D/m$. A weak voter offered less votes no; if any such rejection is certain, both types of H opt out. If all weak offers meet the threshold, proposals below o_0 fail, strict low-only is bounded by the first term of K , and strict pooling is bounded by the second. This completes every deviation at no more than K .

This construction uses the declared freedom of weak PBE at globally off-path proposals. It is not robustly claimed for sequential equilibrium, and it does not follow from weak-vote-passive beliefs alone.

7.2 Exact reduced system

Let h_θ be type θ 's probability of voting yes, and let w_j be weak voter j 's yes probability. If

$$B_H = (1 - \mu)h_0 + \mu h_1 > 0, \quad \nu = \frac{\mu h_1}{B_H}, \quad A = \prod_{j \neq i} w_j, \quad (32)$$

then $A > 0$ requires $x_j \geq c(\nu)$ for every nonproposer, where

$$c(\nu) = \frac{\beta S(\nu)}{m}. \quad (33)$$

Write

$$\mathcal{E} = \sum_{j \neq i} x_j - (m - 1)c(\nu) \geq 0.$$

The current proposer residual and expected payoff are

$$\begin{aligned} r(\nu) &= 1 - y - \delta S(\nu) - \mathcal{E}, \\ \Pi &= B_H [A r(\nu) + (1 - A)c(\nu)]. \end{aligned} \quad (34)$$

The high-type expected IC is

$$\mathcal{D}_1 = Ay + (1 - A)\beta o_1 - o_1. \quad (35)$$

The low-type difference exceeds the high-type difference:

$$\mathcal{D}_0 - \mathcal{D}_1 \geq (o_1 - o_0)\{1 - (1 - A)\beta\} > 0. \quad (36)$$

Therefore $h_1 > 0 \Rightarrow h_0 = 1$. Reverse separation is impossible.

7.3 Regular correspondence

For $\nu \in (0, \mu]$, define

$$B(\nu) = \frac{1 - \mu}{1 - \nu}, \quad h_1(\nu) = \frac{\nu(1 - \mu)}{\mu(1 - \nu)}. \quad (37)$$

Theorem 3 (proved: regular unanimity PBE-UD). Conditional on a terminal equality assessment that attains the R2 value, a regular unanimity PBE-UD exists if and only if

$$\boxed{\bar{y} > o_1 \quad \text{and} \quad \mu > \nu_2^*} \quad (38)$$

or, equivalently, $P > L(\mu)$.

Every equilibrium candidate has $\nu \in (\nu_2^*, \mu]$, so $S(\nu) = P$, $h_0 = 1$, and the tuple $(\nu, A, y, (x_j, w_j)_{j \neq i})$ satisfies:

$$\begin{aligned}
0 < A \leq 1, \quad o_1 < y \leq \bar{y}, \quad x_j \geq \frac{\beta P}{m} \text{ if } w_j > 0, \\
x_j > \frac{\beta P}{m} \Rightarrow w_j = 1, \quad r(\nu) \geq 0, \quad \Pi \geq K(\mu).
\end{aligned} \tag{39}$$

If $\nu < \mu$, the high type mixes, $h_1 = h_1(\nu) \in (0, 1)$, $A \in (0, 1)$, and $\mathcal{D}_1 = 0$. If $\nu = \mu$, both types vote yes and $\mathcal{D}_1 \geq 0$; A may lie anywhere in $(0, 1]$. At least one weak voter who mixes must receive exactly $\beta P/m$. For $N = 3$, there is only one such weak nonproposer, so $A < 1$ implies $\mathcal{E} = 0$.

Proof. Low-only with $h_0 = 1, h_1 = 0$ has maximum

$$\Pi_L(A) = (1 - \mu) [aD - (1 - A)(1 - \beta)]. \tag{40}$$

For $A < 1$ this is strictly below the first component of K ; for $A = 1$, attaining the boundary would require the dominated action $H_0 = Y$ at $y = o_0$. Delay yields only $(1 - \mu)\beta D/m < a(1 - \mu)D$. Thus any equilibrium needs $h_1 > 0$.

For every candidate with $\Pi \geq K$, it must be that $S(\nu) = P$. If $S(\nu) = L(\nu)$, the maximum payoff inside the brackets in (34) is strictly below $aL(\nu)$, while $B(\nu)aL(\nu) = a(1 - \mu)D$, contradicting $\Pi \geq K$. Therefore $\nu > \nu_2^*$, which requires $\mu > \nu_2^*$. The high type's positive participation requires $y > o_1$, which requires $\bar{y} > o_1$.

For sufficiency, set $\nu = \mu, A = 1, \mathcal{E} = 0$, and choose

$$y = o_1 + \eta, \quad 0 < \eta \leq \min\{\bar{y} - o_1, aP - K(\mu)\}. \tag{41}$$

When $P > L(\mu)$, both entries in K are strictly below aP . The proposal gives $\Pi = aP - \eta \geq K$, and Lemma 6 supplies the global completion. $\square$

The theorem includes three on-path forms:

1. **pure pooling with all weak yes:** $\nu = \mu, A = 1$;
2. **pure H pooling with weak mixing:** $\nu = \mu, A \in (0, 1)$;
3. **semi-pooling:** $\nu \in (\nu_2^*, \mu)$, with high-type H mixing.

For high-type mixing, its binding IC gives

$$y_1(A) = \beta o_1 + \frac{(1 - \beta)o_1}{A} > o_1 \tag{42}$$

and, with minimum weak transfers,

$$\Pi_{SP}(\nu, A) = B(\nu) [P - \delta P - (1 - A)\{1 - \beta(o_1 + P)\}]. \tag{43}$$

The feasibility tests are $y_1(A) \leq \bar{y}$, $1 - y_1(A) - \delta P \geq 0$, and $\Pi_{SP} \geq K$. Equation (43) applies when the high type is indifferent, $\mathcal{D}_1 = 0$, including the $\nu = \mu$ endpoint when equality holds. If $\nu = \mu$ and $\mathcal{D}_1 > 0$, the candidate must instead be evaluated directly from (34); relative to the binding formula its payoff is reduced by $A\{y - y_1(A)\}$ and by $A\mathcal{E}$.

7.4 Payoffs, mixing, and selections

For every regular candidate,

$$T_W^U = B(\nu) [A(1 - y) + (1 - A)\beta P] < P, \tag{44}$$

and $V_W^U = T_W^U/m$ under the collective formation convention.

Let $v_H = Ay + (1 - A)\beta o_1$. High-type mixing sets $v_H = o_1$; pure high acceptance requires $v_H \geq o_1$. Since the low type accepts surely,

$$E[u_H^U] \geq o_1. \quad (45)$$

For the T^Y construction (41),

$$\Pi = aP - \eta, \quad T_W^U = P - \eta, \quad E[u_H^U] = o_1 + \eta. \quad (46)$$

A proposer mixture adds no local class: every pure proposal in its support must satisfy (39) and deliver the same proposer payoff. Such mixtures convexify supported outcome vectors.

Under global T^Y , the pure construction is available. Under global T^N , terminal R2 has no best response, so the complete game has no PBE. A proper global T^p likewise fails unless a maximizing terminal boundary is still accepted with probability one. With a valid T^Y terminal continuation but mixing only in R1, the full correspondence in Theorem 3 applies.

The proposer tie-break operates only among actual payoff ties within a fixed completion. It does not rank equilibria sustained by different off-path belief systems. Within a tied support, minimizing H selects the smallest expected value in (45); maximizing H selects the largest; without selection, all tied candidates remain.

Table 4: Regular R1 unanimity survival under PBE-UD.

Former R1-U class	PBE-UD status on the regular domain	Reason
exact low-only $y = o_0$	rejected	low-type yes is dominated on the separating path
exact pooling $y = o_1$	rejected	high-type yes is dominated
deliberate rejection/delay	rejected	proposer payoff lies below the security floor
strict overpaid pooling	survives with a new proof	available iff $\bar{y} > o_1$ and $\mu > \nu_2^*$
semi-pooling / weak mixing	newly required	part of the full correspondence

8 Boundaries and one-sided limits

These are literal boundary games unless explicitly labeled as limits. Equality behavior is part of each statement; no result is filled in by silent continuity. Throughout the literal boundary subsections the prior remains interior, $\mu \in (0, 1)$; prior endpoints are treated only as one-sided limits below.

8.1 $o_0 = 0, \beta < 1$

At $y = o_0 = 0$, low-type implementation, continuation, and opt-out all pay zero. Low-type yes therefore remains admissible. Let

$$K_0 = \begin{cases} \max\{a(1 - \mu), P - \delta\}, & \bar{y} > o_1, \\ a(1 - \mu), & \bar{y} = o_1, \end{cases} \quad (47)$$

so the strict-pooling component is absent, rather than set to zero, at the cap.

Proposition 3 (proved, T^Y). Exact low-only under unanimity gives proposer payoff $a(1 - \mu)$. If $\bar{y} > o_1$, it exists if and only if $a(1 - \mu) \geq P - \delta$; if $\bar{y} = o_1$, it always exists. Strict high-participation equilibria also exist when $\bar{y} > o_1$ and $\mu > o_1$, with Theorem 3's candidate system evaluated at $D = 1$. Thus at least one unanimity PBE-UD exists everywhere on this boundary under T^Y , though the correspondence can be multiple.

For a general fixed equality assessment, the exact low-only class attains $a(1-\mu)$ only when the low type of H and every equality-paid weak voter needed for unanimity vote yes with probability one. It is a proposer best response only if this attained value is no smaller than the active strict-pooling security component $P - \delta$ when $\bar{y} > o_1$. If the required equality actions do not attain the class, strict deviations approach its full value and the partially implemented class cannot be a best response. The strict high-participation candidates are governed by the full system in Theorem 3 with $D = 1$, rather than by an equality convention.

Under majority, exact low-only is also attained. For $N \geq 4$, define

$$B_0 = (1 - \mu)(1 - rc) + \mu c. \quad (48)$$

Let $V_0 = \max\{E, B_0\}$. Under a fixed equality assessment, a full $N \geq 4$ majority PBE-UD exists if and only if: terminal free winning supports are valid; at least one class equal to V_0 is attained; and, when $\bar{y} > o_1$, $V_0 \geq C$. Attaining E requires k threshold-paid sure weak yes votes; attaining B_0 requires the low type of H and the r threshold-paid weak supporters to vote yes surely. Under global T^Y , both classes are attained. For $N = 3$, apply Theorem 2 with B_0 in place of B : use $\max\{E, B_0, C\}$ off the cap and $\max\{E, B_0, Q_1(\lambda_1)\}$ at the cap, always conditional on valid terminal support and the stated attainment test.

8.2 $o_1 = 1, \beta < 1$

Here $\bar{y} = o_1 = 1$, so strict high participation is impossible. If $o_0 > 0$, regular-style low-only remains unattained and unanimity has no PBE-UD. If $o_0 = 0$, exact low-only exists exactly when its low-type and weak equality actions attain the class; in particular it exists under T^Y .

Conditional on a valid terminal support, majority pooling is payoff-irrelevant. At $N = 3$, proposer value is $\max\{E, B\}$ provided a maximizing class is attained as in Theorem 2. At $N \geq 4$ and $0 < o_0 < 1$, existence requires $E \geq B$ and an attained exclusion support, and the outcome is exclusion. The corner $o_0 = 0, o_1 = 1$ is governed by the preceding subsection, where exact low-only may also attain the maximum.

8.3 $\beta = 1$

At no discounting, a failure continuation can pay each type exactly its outside option. Exact H thresholds again remain admissible, but a class that relies on equality votes is available only when those votes actually attain it. The formulas below first identify the candidate values and then separate the full attainment statement from the global T^Y selection.

For unanimity, set

$$K_1 = \max \left\{ \frac{L(\mu)}{m}, P - \frac{m-1}{m} D \right\}. \quad (49)$$

Under global T^Y , if $L(\mu) > P$, exact low-only and deliberate continuation give proposer payoff $L(\mu)/m$. If $P > L(\mu)$, exact pooling and deliberate continuation give P/m . At $L(\mu) = P$, exact low-only, exact pooling, and deliberate continuation all enter the unselected correspondence when their required equality actions are attained. Exact low-only also survives in the second region when

$$P - \frac{m-1}{m} D \leq \frac{L(\mu)}{m}. \quad (50)$$

For a general fixed equality assessment, every exact low-only, exact pooling, or delay class must be attained by all equality actions it requires; the proposer can mix only among attained pure classes that share the maximum. Strictly overpaid pooling is evaluated from the general system (34). A payoff strictly above K_1 is sufficient; equality with K_1 can also remain in the unselected correspondence if it attains the common maximum. Under a minimum- H proposer tie-break, such an equality candidate survives only if it also minimizes H 's payoff among tied best responses.

For majority, use $c = 1/m$. Under T^Y , the exact attained proposer value is

$$\max \{E, (1 - \mu)(1 - o_0 - rc) + \mu c, E - (1 - \mu)o_0, 1 - o_1 - rc\}, \quad (51)$$

over feasible classes. For a general fixed equality assessment, a full PBE-UD exists if and only if terminal free winning supports are valid and at least one class attaining the largest feasible value in (51) is implemented by all equality actions it requires. Proposer mixing is only over attained maximizing classes. At $N = 3$ and $\bar{y} = o_1$, replace the pooling entry by the attained cap value $Q_1(\lambda_1)$ from (27). Under global T^Y , all required exact classes are attained. Delay has value at most c , which can tie E only for $N = 3, 4$. A global T^N still has no full equilibrium because terminal majority lacks a proposer best response; any R1-no sensitivity with a valid terminal T^Y must be labeled hybrid.

8.4 Priors as one-sided limits

Literal $\mu = 0, 1$ games require a convention for zero-probability types. The objects here are cluster limits from $\mu \in (0, 1)$.

For regular unanimity, $\nu_2^* > 0$, so no equilibrium sequence reaches $\mu \rightarrow 0^+$. As $\mu \rightarrow 1^-$, pure-pooling limits exist when $\bar{y} > o_1$. For any fixed

$$0 < \eta \leq \min \{\bar{y} - o_1, aP - \max(0, P - \delta D)\},$$

the limiting vector is

$$\left(aP - \eta, P - \eta, \frac{P - \eta}{m}, o_1 + \eta \right). \quad (52)$$

Semi-pooling can add cluster points; no uniqueness claim is made.

For $N \geq 4$, regular majority exclusion has a lower-endpoint sequence only if $o_0 \geq c$ and, when $\bar{y} > o_1$, $o_1 \geq c$. An upper-endpoint sequence requires only the latter capacity-dependent pooling condition. $N = 3$ under T^Y has the direct limits of $\max\{E, B, C\}$. These are limits of regular equilibria, not solutions of new degenerate games.

9 Entry and institutional comparison

Let

$$\bar{o} = (1 - \mu)o_0 + \mu o_1. \quad (53)$$

The regular common domain consists of (38), a valid terminal equality support, and the applicable majority existence condition: Theorem 1 for $N \geq 4$, or the attainment condition in Theorem 2 for $N = 3$. Because regular unanimity requires $\bar{y} > o_1$, the $N = 3$ cap class Q_1 lies outside this common regular domain.

Proposition 4 (proved: regular entry nesting). In every regular unanimity PBE-UD,

$$T_W^U < P. \quad (54)$$

In every regular majority PBE-UD on the common domain,

$$T_W^M \geq P. \quad (55)$$

Therefore

$$V_W^U < V_W^M, \quad \mathcal{F}_U \subsetneq \mathcal{F}_M. \quad (56)$$

Proof. Equation (44) is a probability-weighted combination of $1 - y < P$ and $\beta P < P$, multiplied by $B(\nu) \leq 1$. For $N \geq 4$, majority exclusion gives total one. For $N = 3$, an attained exclusion maximizer has $E \geq C = P$; an attained low-only maximizer has $B \geq C = P$, and its total weak payoff is $B + \mu c \geq P$; pooling gives exactly P . Mixtures preserve the inequality. Since the unanimity inequality is strict, the interval $(V_W^U, V_W^M]$ is nonempty and establishes the proper formation-set inclusion in (56). $\square$

For a selected pair $(v_U, v_M) = (V_W^U, V_W^M)$:

1. both institutions form if $\chi \leq v_U$;
2. only majority forms if $v_U < \chi \leq v_M$;
3. neither forms if $\chi > v_M$.

There is no regular common-domain region in which only unanimity forms. Unlike the old scalar result, v_U is generally a correspondence because strict pooling, weak mixing, and semi-pooling can coexist.

Proposition 5 (proved: H comparison). In every regular majority PBE-UD,

$$\bar{o} \leq E[u_H^M] \leq o_1. \quad (57)$$

In every regular unanimity PBE-UD,

$$E[u_H^U] \geq o_1. \quad (58)$$

Hence unanimity weakly benefits H whenever both rules form on the common domain. Equality is possible only in the $N = 3$ majority pooling class when unanimity keeps the high type exactly indifferent; the T^Y strict-pooling construction gives a strict advantage $o_1 + \eta > o_1$.

The mechanism therefore survives, but with new mathematics: exact pooling at o_1 is removed, strict overpayment or mixing replaces it, and majority exclusion becomes the unique regular payoff for $N \geq 4$ whenever it exists.

9.1 Comparative statics

- Unanimity existence is monotone in the prior: it begins strictly above ν_2^* , and it requires positive capacity slack $\bar{y} - o_1$.
- Under the T^Y pure construction, greater η transfers surplus one-for-one from weak states to H ; the admissible upper bound is $\min\{\bar{y} - o_1, aP - K\}$.
- For majority with $N \geq 4$, if $o_0 < c$, increasing μ past μ_M removes the low-only nonattainment threat. If $o_0 \geq c$, that threat never binds. When capacity is slack, $o_1 \geq c$ removes the pooling threat.
- Increasing β raises the weak approval price $c = \beta/m$ and can shrink majority's existence domain. Boundary $\beta = 1$ is not obtained by simply substituting into the regular theorem because exact H thresholds re-enter.

10 Reproducible verification

All computation is in new, separate R scripts:

- **verify_undominated_voting_gate0.R**: finite ballot checks of the local dominance lemmas, the $N = 3$ exception, and the pooling-continuation counterexample;
- **verify_undominated_voting_regular.R**: majority support enumeration, regular existence identities, $N = 5, q = 3$, and three unanimity witnesses (strict pooling, pure- H weak mixing, and semi-pooling);
- **verify_undominated_voting_boundaries.R**: $o_0 = 0$, $o_1 = 1$, $\beta = 1$, and one-sided-limit algebra, with every row labeled T^Y where applicable;
- **verify_undominated_voting_all.R**: orchestration wrapper.

The current generated checks report:

Table 5: Reproducible Goal 3 verification summary.

Verifier	Result	Evidentiary status
Gate 0 finite ballots	116/116 PASS	checked numerically
Regular-domain identities and enumeration	6629/6629 PASS	checked numerically
Boundary identities	1488/1488 PASS	checked numerically

The CSV outputs under **tables/** record parameters, source, date, and per-check status. These checks verify implementation of the displayed identities; the general results rest on the analytical proofs above.

11 Result ledger

Table 6: Goal 3 formal result ledger.

Object	Final status	Goal 3 conclusion
Simultaneous sealed ballot	proved	preserved; no order of H exists
Terminal U value $S(\nu)$	proved	survives only conditional on boundary attainment
Terminal M payoff	proved	unique conditional on endogenous free-support existence
Exact R1-U low-only	rejected	dominated H_0 -yes on separating path
Exact R1-U pooling	rejected	dominated H_1 -yes
Strict/semi-pooling under U	proved	full correspondence in Theorem 3
Old regular U existence condition	rejected	replaced by $\bar{y} > o_1, \mu > \nu_2^*$
Old majority floor F_M	rejected	replaced by attained E and nonattained B, C
Old large- N interval $[F_M, 1]$	rejected	regular $N \geq 4$ payoff collapses to exclusion
Old No-Cheap-H condition	rejected	replaced by (25) and (24)
Entry nesting	proved anew	strict on the regular common domain
Informational advantage for H	proved anew	$E[u_H^U] \geq o_1 \geq E[u_H^M]$
Global T^N equilibrium	rejected	terminal proposer best response absent

12 Scope and limitations

The regular unanimity sufficiency proof is a weak-PBE construction. It uses off-path proposal beliefs that are not linked by consistency to nearby on-path proposals. No claim is made for sequential equilibrium, trembling hand perfection, coalition-proofness, endogenous rule choice, or $\pi_H > 0$. Equality selections are reported rather than silently fixed.

Most importantly, none of these results uses public or sequential voting inside a ballot. Later voters never observe earlier votes because there are no later voters within the simultaneous ballot. Baron–Ferejohn motivates excluding weakly dominated voting actions; it does not supply this paper’s protocol.